

HR Analytics Case Study

Workforce Intelligence Report — IBM Dataset

Dataset	IBM HR Analytics Employee Dataset
Employees	1,470 Records 35 Variables
Tools Used	Microsoft Excel Pivot Tables Dashboards
Analysis	Salary Performance Hiring Attrition
Date	June 2026

01. Executive Summary

This case study presents a comprehensive analysis of IBM HR dataset comprising 1,470 employees across three departments: Human Resources, Research and Development, and Sales. The objective is to identify key workforce trends across salary distribution, employee performance, hiring demographics, and attrition patterns.

Using Microsoft Excel — including Pivot Tables, calculated columns, and interactive dashboards — this analysis uncovers actionable insights that HR managers and business leaders can use to improve employee retention, optimize compensation, and strengthen workforce planning.

Key Performance Indicators

1,470 TOTAL EMPLOYEES 3 Departments	Rs.6,503 AVG MONTHLY SALARY Rs.1K - Rs.20K	16.1% ATTRITION RATE 237 Employees Left	3.15 AVG PERFORMANCE Scale: 1 - 4
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Scope of Analysis

- Salary Analysis — Average compensation by department and job role
- Performance Distribution — Employee rating breakdown across departments
- Hiring Trends — Age group distribution of the workforce
- Gender Pay Analysis — Compensation equity across job levels
- Attrition Analysis — Impact of overtime on employee turnover

02. Key Findings

#	Finding	Metric	Impact
1	Sales has highest avg salary among all departments	Rs.6,959	Sales compensation reflects strong commission structure
2	Majority of workforce is in 26-35 age group	606 / 1,470	Young workforce — high energy but higher attrition risk
3	84.6% rated Good, only 15.4% rated Excellent	1,244 : 226	Limited top performers — performance system needs revision
4	Overtime employees have higher attrition rate	127 vs 110	Burnout risk is significant — overtime policy needs review
5	Gender pay gap is minimal across all job levels	Less than 2%	Strong pay equity — positive employer branding signal
6	Mid salary band has highest employee concentration	519 employees	Growth opportunity exists for senior role promotions

Deep Dive Insights

Rs. Salary by Department	Sales leads with Rs.6,959 avg monthly income, followed by HR at Rs.6,655 and R&D; at Rs.6,281. Despite R&D; having the largest workforce (961 employees), their salary is lowest — suggesting junior-heavy team composition.
* Performance Gap	Only two active performance ratings exist (3=Good, 4=Excellent). 84.6% rated Good suggests limited differentiation in performance reviews — a revised 5-tier rating system would provide better talent visibility.
A Age Distribution	The 26-35 age band dominates with 606 employees (41.2%). The 46+ segment has 273 employees — succession planning for senior roles should be a strategic priority for HR leadership.
! Overtime-Attrition Link	Overtime employees showed 15% higher attrition (127 vs 110). This strong correlation suggests overtime is a key burnout driver — HR should consider workload redistribution and strict overtime caps.

03. Strategic Recommendations

1

Revise Overtime Policy

Implement a strict overtime cap of 10 hrs/week. Introduce compensatory offs and monitor overtime employees quarterly to prevent burnout and reduce the 15% higher attrition risk identified in this analysis.

2

R&D; Compensation Review

Research and Development has the largest workforce yet lowest avg salary. A structured salary revision for mid-level R&D; roles could improve retention and attract senior talent to this critical department.

3

Strengthen Performance Differentiation

With 84.6% rated Good and only 15.4% Excellent, the review system lacks differentiation. A 5-tier rating system with clear KPIs will help identify and reward top performers effectively.

4

Succession Planning for 46+ Workforce

273 employees are 46+ years. Proactive succession planning, knowledge transfer programs, and identifying high-potential employees in the 26-35 group will ensure business continuity.

04. Conclusion

This HR Analytics case study demonstrates the power of data-driven decision making in workforce management. By analyzing 1,470 employee records across salary, performance, demographics, and attrition dimensions, four critical action areas were identified for HR leadership.

The most urgent finding is the strong correlation between overtime and attrition — addressing this alone could significantly reduce turnover costs. R&D; compensation restructuring and performance system enhancement offer additional high-ROI improvement opportunities.

Tools and Techniques Used

- Microsoft Excel — Data cleaning, calculated columns, formulas
- Pivot Tables — Multi-dimensional data aggregation and analysis
- IF / COUNTIF / AVERAGE — Custom segmentation formulas
- Interactive Dashboard — KPI cards, 5 charts, dark theme
- IBM HR Dataset — 1,470 rows, 35 columns (Source: Kaggle)